# DAVID RAJ J

**AI Engineer • Generative AI • Agentic Systems • RAG**

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## PROFESSIONAL SUMMARY

AI Engineer with 3+ years building production Generative AI systems — agentic workflows, Retrieval-Augmented Generation (RAG), and multi-tool orchestration. Specialized in designing custom **MCP (Model Context Protocol)** servers and integrating them into **LangGraph** agents for secure, context-aware tool calling across Gmail, Google Workspace, Dropbox, and enterprise APIs. Proven record delivering scalable async FastAPI backends and citation-backed RAG pipelines with measurable gains in retrieval precision (up to 38%), latency (up to 42%), and analyst efficiency (up to 40%). Skilled across LangChain, LangGraph, vector databases, and LLM deployment on AWS.

## TECHNICAL SKILLS

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|---|---|---|---|
| **Generative AI & LLM** | RAG, LangChain, LangGraph, MCP, ReAct Agents, Tool / Function Calling, Prompt Engineering, LLM Fine-tuning, Transformers, NLP | **Backend & APIs** | FastAPI, Flask, Async Python, REST, Webhooks, WebSocket / SSE, SQLAlchemy 2.0, Pydantic v2 |
| **Languages & Core** | Python, SQL, Linux | **Vector & Databases** | ChromaDB, Pinecone, Qdrant, MongoDB, PostgreSQL |
| **ML & Deep Learning** | TensorFlow, PyTorch, Supervised & Unsupervised Learning, LSTM, EDA, Statistical Analysis | **Cloud & MLOps** | AWS (Bedrock, SageMaker, ECR, ECS), Docker, MLflow, Alembic, CI/CD |

## PROFESSIONAL EXPERIENCE

### Contus Tech — AI Engineer
**Jun 2025 – Present**

*Chennai, India*

- Designed and deployed custom MCP servers enabling secure, tool-based agent interactions with Gmail, Google Calendar, Google Drive, Dropbox, internal enterprise APIs, and external STT/TTS voice-pipeline services.
- Integrated MCP tools directly into LangGraph agent nodes for dynamic, context-aware tool calling, voice-agent communication, and multi-step automated workflows.
- Built webhook-driven action tools using LLM function-calling to extract structured parameters and trigger backend API workflows across enterprise systems.
- Engineered a highly scalable async FastAPI backend serving high volumes of concurrent LLM, MCP, and distributed voice-agent requests.
- Designed an enterprise RAG workflow delivering citation-backed responses, improving retrieval precision by 38%.
- Optimized vector-database performance (ChromaDB, Pinecone), reducing query latency by 42% and doubling throughput.

### Wellspring Systems Pvt Ltd — ML Engineer
**Mar 2023 – May 2025**

*Chennai, India*

- Built autonomous GenAI agents for phishing and XSS detection using LSTM neural networks, achieving 89% accuracy and reducing false positives by 20%.
- Developed a LangGraph-powered RAG system integrating cybersecurity playbooks, standardizing incident handling across 10+ scenarios and improving analyst response time.
- Created LLM-powered summarization and analytics tools enriched with MITRE ATT&CK, CVE, and TTP context, reducing manual reporting effort by 40%.
- Implemented an LLM threat-reasoning engine for automated alert correlation and recommended actions, lowering mean time to investigate (MTTI) by 35%.

## KEY PROJECTS

**AgentLens** — Agentic AI Control Panel & Observability Console **2026**

**Stack:** Python 3.12, FastAPI, MongoDB (Motor), Pydantic v2, Next.js, TypeScript, Tailwind, SSE

- Built a control panel to configure and observe a tool-using LLM agent, rendering each run as a live step-by-step timeline of reasoning → tool calls → retrievals → final answer over Server-Sent Events.
- Implemented a RAG knowledge-base pipeline: PDF upload → parse → chunk → embed into datasets the agent retrieves from at query time.
- Added MCP tool integration (discover and toggle MCP server tools) plus custom HTTP webhook tools with schema validation and an SSRF-guarded live-execution mode.
- Enforced a frozen UI⇄API contract with ULID entity IDs, a single unified SSE event vocabulary, and one stream consumer driving the agent timeline.

**FlowSphere** — Visual Workflow-Automation Platform with AI Copilot **2026**

**Stack:** Python 3.12, FastAPI, Async SQLAlchemy 2, PostgreSQL, Next.js 16, TypeScript, React Flow, APScheduler, WebSockets

- Engineered an n8n/Zapier-class automation engine — a node-based execution engine running user-defined workflows with schedule, webhook, and wait triggers.
- Built a safe expression evaluator (a `{{ }}` Python subset over an AST allow-list) and an item-based execution model passing typed data between node ports.
- Implemented an AI copilot that proposes staged, server-validated graph diffs without mutating saved workflows — node types validated against a registry, LLM provider injected via config.
- Delivered realtime run observability over WebSockets and Fernet-encrypted, write-only credential storage behind a standardized error envelope.

**Loop** — Multi-Tenant AI Voice-Feedback SaaS **2026**

**Stack:** Python, FastAPI, Async SQLAlchemy 2.0, PostgreSQL, Alembic, JWT, Pydantic v2, Docker

- Architected a multi-tenant SaaS backend where an AI voice agent auto-calls customers post-service, records and transcribes calls, and converts them into structured insight (ratings, sentiment, NPS).
- Designed a fully async FastAPI + SQLAlchemy 2.0 (asyncpg) service layer with tenant-scoped data isolation, JWT auth, and role-based access control (owner/admin/agent/viewer).
- Modeled 13+ multi-tenant PostgreSQL entities with UUID keys, Alembic migrations, and a reusable generic async CRUD base service.
- Integrated telephony, speech-to-text, and TTS providers (Twilio, Deepgram, ElevenLabs) through a pluggable integration layer feeding campaign-driven call flows.

## EDUCATION

**Scaler Academy** — Advanced Data Science & Machine Learning **Nov 2023 – Jun 2025**

**Guvi Geek Network (IIT-M)** — Advanced Programming Professional & Master Data Science **Jun 2021 – Nov 2022**

**Saveetha Engineering College** — B.E., Electrical & Electronics Engineering **Aug 2012 – Apr 2016**

## LANGUAGES

**English • Tamil**